# Rules or Learned Patterns?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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TEACHER GUIDE · 1 OF 9

## Rules or Learned Patterns?

> Prepare Rules or Learned Patterns?: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Execute a stated rule including its boundary

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a measurement using its quantity, value, unit and conditions

Reference example, not the only acceptable wording: Keep the observed value 24 and approximate time lunchtime; mark the unit, exact time if needed, location, exposure and device identity as unknown or to be checked.

A suitable completed record needs air temperature, value, unit and relevant conditions. Invented degrees C or shade is not an **observation**.

### AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an observed result from an explanation of how it was produced

Reference example, not the only acceptable wording: **Observation**s include that the door opened when the visitor approached and that its **label** says intelligent entrance.

A useful question asks whether a written sensor **threshold**, a model fitted on examples or a person controls opening. The **label** is an observed marketing claim, not **mechanism evidence**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

> Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

### Concept to explain

Fixed instructions and learned behavior can look similar but arise differently.

| Term | Meaning |
| --- | --- |
| fixed rule | instruction written directly in advance |
| learned pattern | relationship fitted from examples |
| training | process that adjusts a model from examples |
| inference | using the finished model on a new input |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

## Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

This is a genuine small **fitting** procedure: choose among candidate **threshold**s by empirical disagreement count. It is intentionally executable on paper.

Do not equate all AI with machine learning or all hand**written rule**s with non-AI; the lesson isolates **fitting** versus execution.

The fictional **label** convention avoids implying that size words have universal numerical definitions.

### Readiness answer

The warning stays off at exactly 28. Check that the reading is the intended quantity in degrees C and is relevant to the time and conditions.

Width and length cards with **label**s on a separate sheet

Two candidate-**threshold** cards for each exercise

A trace-table frame with **input**, **prediction**, supplied **label** and disagreement columns

### End goal for the pictorial case

By the end, I can distinguish choosing a rule from applying an already chosen rule.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Rules or Learned Patterns? using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

### 1. Input — information entering

**INPUT** a motion value or image values

### 2. Process — the deciding step

**PROCESS** run an explicit if/then rule or a learned classifier

### 3. Output — the result

**OUTPUT** turn on a light or predict an animal **label**

### Key idea

CORE IDEA Fixed instructions and learned behavior can look similar but arise differently.

### How the mechanism works

UNDER THE HOOD A fixed program's decision logic is specified directly by people. A **learned model**'s parameters are fitted from examples.

Both are software, both may be complex, and both can fail; the distinction is how the decision behavior was produced.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

### The case to investigate

CASE A motion light and an animal recognizer both react to **input**, but one checks a written condition while the other uses patterns learned from examples.

### Step 1

STEP 1 — OBSERVE a motion value or image values

### Step 2

STEP 2 — TRACE run an explicit if/then rule or a learned classifier

### Step 3

STEP 3 — CONCLUDE turn on a light or predict an animal **label**

### Model answer

MODEL REASONING The motion light follows a directly written sensor condition. The animal recogniser uses a model fitted on **label**led examples.

Observing that both react to **input** does not tell us how their decisions were produced.

### A claim to avoid

DO NOT OVERCLAIM Do not use complicated versus simple as a substitute for fixed versus learned.

What new **evidence** could change this answer?

### Teacher precision note

This course concentrates on modern machine learning. AI also includes symbolic and rule-based approaches. A simple automatic switch or robot body is not, by itself, **evidence** of AI.

Discuss how the system reasons or learns, rather than judging appearance.

### Concrete teaching case

#### The teaching case

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

How were their decision **process**es obtained? Where should A send blue cards?

### Exact explanation to model

#### Why the result follows

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

### Explain the added worked example

## Explain the added worked example · 1

A paper sorting task describes imaginary cards. Each card has a width in centimetres and a recorded **label**, narrow or wide. One person writes a rule from a design requirement: **label** a card wide when its width is at least 5 cm.

Another person compares several candidate **threshold**s with **label**led examples and chooses the one with fewest disagreements. Both may end up using the same **threshold**, but they selected it in different ways.

Executing a rule means applying it to an **input**. **Fitting** means using examples to select or adjust the rule or model. A fitted model can be very simple.

We do not need a brain-like picture or a complicated computer to study the difference between selecting a rule and using it.

In this exercise the **label**s are supplied by the task designer. They are not a universal definition of wide, and a centimetre value alone would not determine every possible meaning of that word.

State what the recorded **label**s mean in the task before counting disagreements.

For **threshold** t, our rule predicts wide when width is at least t and narrow otherwise. At least includes equality: with t=5, width 5 is wide. Changing at least to greater than changes the rule and may change the **error count**.

A **fitting** procedure also contains choices people made: the **feature**, candidate **threshold**s, **label** definition and way of counting errors. Calling a **threshold** learned does not remove human responsibility for those choices.

### Key terms

| Term | Meaning |
|---|---|
| Feature | A recorded property used as an input to a decision or model |
| Label | A recorded category or target associated with an example |
| Threshold | A boundary value used in a comparison |
| Fitting | Choosing or adjusting a model using examples |
| Error count | The number of predictions that disagree with the comparison labels |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Width in cm | Supplied label | Threshold 3 prediction | Threshold 5 prediction |
|---|---|---|---|
| 2 | narrow | narrow | narrow |
| 4 | narrow | wide | narrow |
| 6 | wide | wide | wide |
| 8 | wide | wide | wide |

## Model the reasoning aloud · 2

Apply **threshold** 3 to each width before checking the **label**. Only width 4 disagrees, so its **error count** is 1. **Threshold** 5 agrees on all four examples, so its **error count** is 0. Of these two candidates, this procedure selects 5.

Zero disagreements here means it fits these four **label**s; it is not a guarantee about all future cards.

## Model the reasoning aloud · 3

There can be ties. **Threshold** 6 also agrees on these four examples because 6 is included by at least. **Threshold**s 5 and 6 predict differently for width 5, which is absent from the **fitting** set.

The available examples therefore do not uniquely determine how that width should be **label**led. A procedure needs an explicit tie rule, and we should record the unresolved case.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Rules or Learned Patterns?

A fictional paper sorter learns a size boundary for toy pebbles.

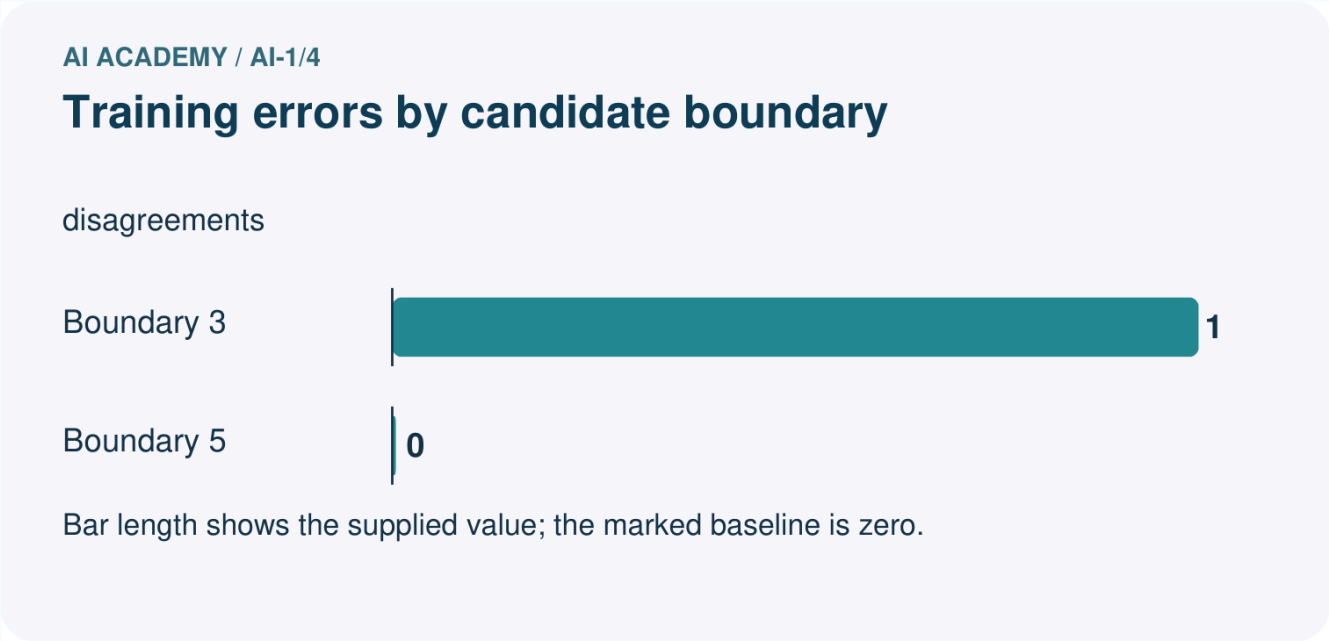
### The facts we will use

#### Supplied facts

**Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large.

#### Supplied facts

Compare large-at-least-3 with large-at-least-5; choose fewer errors.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/4

# Rules or Learned Patterns?

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Training examples

Diameters 1,3,5,7 have reference labels S,S,L,L.



### 2. Candidate 3

Predictions S,L,L,L: one disagreement at 3.



### 3. Candidate 5

Predictions S,S,L,L: zero disagreements.



### 4. New pebble

Selected boundary 5 predicts small for a new diameter 4.

**CHANGE ONE THING**

Give a new pebble diameter 5 instead of 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Training examples | Diameters 1,3,5,7 have reference labels S,S,L,L. |
| 2. Candidate 3 | Predictions S,L,L,L: one disagreement at 3. |
| 3. Candidate 5 | Predictions S,S,L,L: zero disagreements. |
| 4. New pebble | Selected boundary 5 predicts small for a new diameter 4. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows


### Why the result follows


The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

### What this example does not establish

#### Keep the stated limit

Zero **training** disagreements do not prove every future pebble receives the correct **label**.

### Change one thing in the pictured case

#### Change one thing

Change: Give a new pebble diameter 5 instead of 4.

#### Predicted result

**Prediction**: The selected rule predicts large.

#### Why the result follows

Reason: At least includes equality, so 5 meets the boundary.

### Ask while showing the picture

## Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Rules or Learned Patterns?, reduce support deliberately and check individual reasoning before discussing answers.

**Board sequence**

### 1. Put the facts on the board

Write the given facts: Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

### 2. State the decision to explain

Write the question to explain: How were their decision **process**es obtained? Where should A send blue cards?

### 3. Reveal the worked reasoning

Reveal after **prediction**s: A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

### Hold back the independent task

Keep for an independent attempt: System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

For widths 1,3,5,7, **threshold** 3 predicts narrow,wide,wide,wide and has one error; **threshold** 5 predicts narrow,narrow,wide,wide and has zero.

The choice establishes a lower **fitting** error among these candidates on these examples.

The exit distinction is selecting a parameter using **label**led examples versus keeping the parameter fixed while making a **prediction** on a new **input**.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1 start 0 to 7 minutes | Retrieve the equality boundary from the temperature rule. Compare greater than with at least. |
| 7 to 20 minutes | Contrast a requirement-selected threshold with an example-selected threshold. Read the fitting and use diagram. |

| Stage | Teacher action |
| --- | --- |
| 20 to 33 minutes | Work through all rows for candidates 3 and 5. Count errors only after writing predictions. |
| 33 to 45 minutes | Pairs complete W1 and W2. Ask each learner to explain where the fitting decision occurs. |
| Session 2 start 0 to 8 minutes | Show the tie between 5 and 6. Ask for an input that makes them disagree. |
| 8 to 23 minutes | Complete W3 and discuss what the fitting data does not determine. |
| 23 to 38 minutes | Collect W4 independently with comparison labels for the new case hidden. |
| 38 to 45 minutes | Discuss W5 and the exit explanation. Save the selected rule and its known limitations. |

### Reduce help deliberately

#### Completion practice

Completion task: Widths 1, 3, 5, 7 have **label**s narrow, narrow, wide, wide. For candidates 3 and 5, list all **prediction**s using wide when width is at least the candidate. Count disagreements and select the lower-error candidate.

#### Reasoning to listen for

Teacher check: **Threshold** 3: narrow, wide, wide, wide; one disagreement at 3. **Threshold** 5: narrow, narrow, wide, wide; zero disagreements. Select 5 among these candidates based on these examples only.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: The selection procedure can fit a single **threshold**. Complexity is not the definition of **fitting**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Show which stage used **label**s and which stage used only the new measurement.

### Reasoned teacher answer

Selection used all four **training label**s. The new **prediction** compares diameter with 5 without seeing its reference **label**.

Changed-case reasoning: The selected rule predicts large. At least includes equality, so 5 meets the boundary.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

# Run the weekly evidence lab

> Run the Rules or Learned Patterns? investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

### 1. Predict

State your **prediction** before the result is known.

### 2. Investigate

Complete the task: Compare a motion light with an animal recognizer.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

#### Find the unsupported claim

CLAIM TO REPAIR Every rule must have been learned from data.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A learned rule cannot be written simply: The selection procedure can fit a single **threshold**. Complexity is not the definition of **fitting**.

Zero **fitting** errors means the model is correct everywhere: Use **threshold**s 5 and 6, then **input** 5. Identical **fitting** scores leave different unseen **prediction**s.

The **label** is available when predicting: Cover the new case's **label** until the **prediction** is recorded. Separate information available at **fitting** time from **prediction** time.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Rules or Learned Patterns? questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

A was explicitly programmed; B learned parameters from examples. Blue-card handling is unspecified, so request a rule. Both systems can fail.

### Independent question

#### Independent question · withhold the key

System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

### Expected independent answer

#### Teacher key · Expected independent answer

A's designer writes the rule. **Training** examples can change B's learned behaviour. A new set of examples does not automatically improve B; test it on separate examples.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification **threshold**.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every rule must have been learned from data.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

No. A follows supplied instructions but purple handling is unspecified; B applies a **learned pattern**. Inspect A's complete rule and B's **training** and **prediction process**.

The same **output** can arise from different **mechanism**s and neither result establishes correctness without a purple-card policy.

### Explanations for the additional workbook examples

### W1 Execute before fitting

#### Workbook question · W1

Use wide when width is at least 5 cm. Predict the **label** for widths 4, 5 and 6. Then state what changes if the rule uses greater than 5 instead.

#### Teacher reasoning · W1

At least 5 gives narrow, wide, wide. Greater than 5 gives narrow, narrow, wide. Only the equality case changes among these **input**s.

### W2 Complete the supported comparison

#### Workbook question · W2

Widths 1, 3, 5, 7 have **label**s narrow, narrow, wide, wide. For candidates 3 and 5, list all **prediction**s using wide when width is at least the candidate. Count disagreements and select the lower-error candidate.

#### Teacher reasoning · W2

**Threshold** 3: narrow, wide, wide, wide; one disagreement at 3. **Threshold** 5: narrow, narrow, wide, wide; zero disagreements. Select 5 among these candidates based on these examples only.

### W3 Make a tie visible

#### Workbook question · W3

For the worked widths 2, 4, 6, 8 with **label**s narrow, narrow, wide, wide, compare **threshold**s 5 and 6. Give a width where they predict differently. Explain why the four **fitting** examples cannot decide that case.

#### Teacher reasoning · W3

Both **threshold**s have zero disagreements. At width 5, **threshold** 5 predicts wide while **threshold** 6 predicts narrow.

No **fitting** example is width 5 or otherwise distinguishes these **threshold**s; their equal **fitting** score does not establish a unique answer there.

## W4 Fit a new paper classifier

### Workbook question · W4

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

### Teacher reasoning · W4

**Threshold** 5 predicts short, long, long, long: one error at 5. **Threshold** 7 predicts short, short, long, long: zero errors. Choose 7 and predict short for 6.

This fits these supplied **label**s among these two candidates; the new **label**, **label**s at other lengths and general performance are not established.

## W5 Keep the check honest

### Workbook question · W5

A learner sees that their final-check card was predicted incorrectly, changes the **threshold** to match it and reports the same card as an unseen test. Explain the problem and propose a valid next check.

### Teacher reasoning · W5

The revealed check **label** influenced the revised **threshold**, so it became part of development. Record the revision and use a different, unrevealed set after fixing the rule.

Do not report the adapted-to case as an independent final test.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

### Selected case answer

#### Teacher key · Selected case answer

**Threshold** 5 predicts short, long, long, long: one error at 5. **Threshold** 7 predicts short, short, long, long: zero errors. Choose 7 and predict short for 6.

This fits these supplied **label**s among these two candidates; the new **label**, **label**s at other lengths and general performance are not established.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Rules or Learned Patterns? to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

### Helpful Classroom Sorter

Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 5: Pictures Become Numbers. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Use a number line with a closed dot at the **threshold** and arrows to the long or wide region. Permit oral explanation with a completed **prediction** table.

Extension: ask learners to find every whole-number **threshold** tied for zero errors in the worked example and explain what a tie-breaking rule does.

W4 is secure when both **prediction** lists and counts are correct, **threshold** 7 is selected, length 6 is predicted short and the learner limits the claim to available **evidence**.

An arithmetic correction alone is insufficient if the **fitting** versus **prediction** distinction is missing. Retest with different lengths and **threshold**s after reteaching.

Week 5 asks how an image becomes numerical **input** and what spatial information must be preserved before a model can use it.

### Plan the delayed return

Suggested return points in this level: ai-1/5, ai-1/7, ai-1/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Rules or Learned Patterns? lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Execute a stated rule including its boundary

Prepare the **label**led worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Widths 1, 3, 5, 7 have **label**s narrow, narrow, wide, wide. For candidates 3 and 5, list all **prediction**s using wide when width is at least the candidate.

Count disagreements and select the lower-error candidate.

### Supported practice answer

Reasoned practice answer: **Threshold** 3: narrow, wide, wide, wide; one disagreement at 3. **Threshold** 5: narrow, narrow, wide, wide; zero disagreements. Select 5 among these candidates based on these examples only.

Check every learner before discussion: Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: The selection procedure can fit a single **threshold**. Complexity is not the definition of **fitting**.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the **label**led picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Give a new pebble diameter 5 instead of 4. Require a **prediction** and a reason before comparing. Expected result: The selected rule predicts large. Reason: At least includes equality, so 5 meets the boundary.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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TEACHER GUIDE · 8 OF 9

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

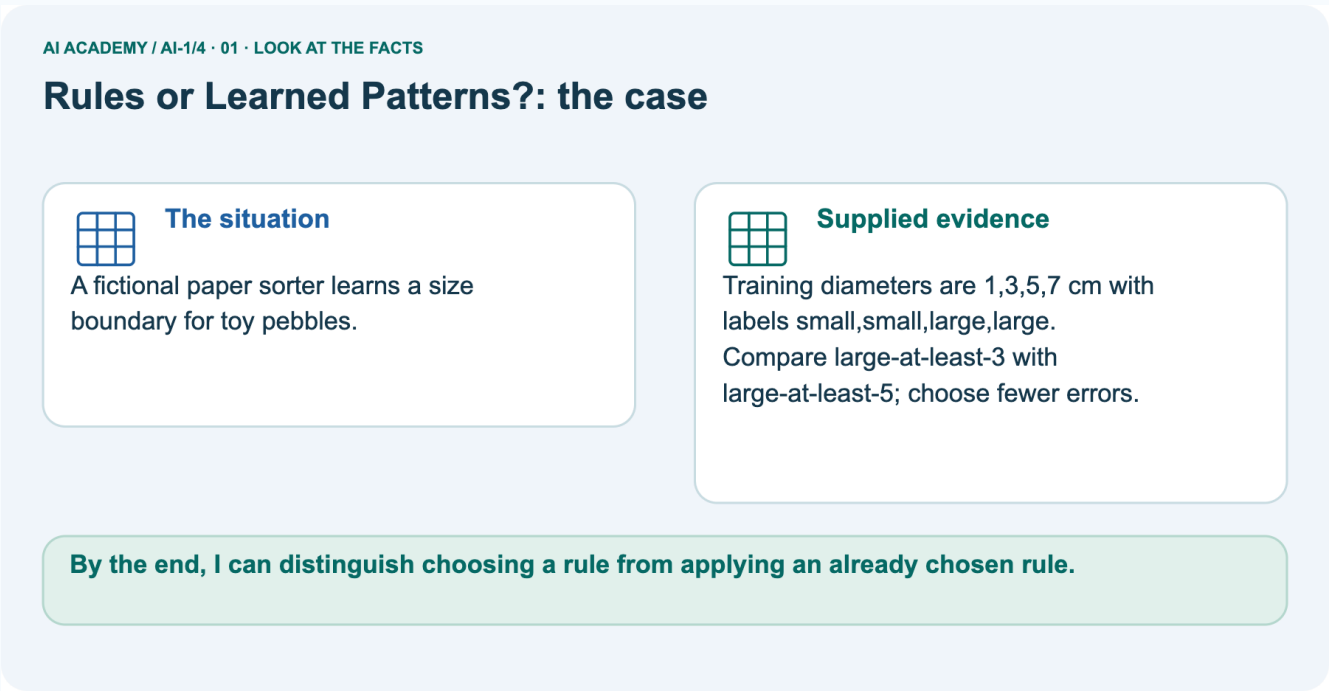
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

# Visual concept journey

> Show one Rules or Learned Patterns? graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Rules or Learned Patterns?: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

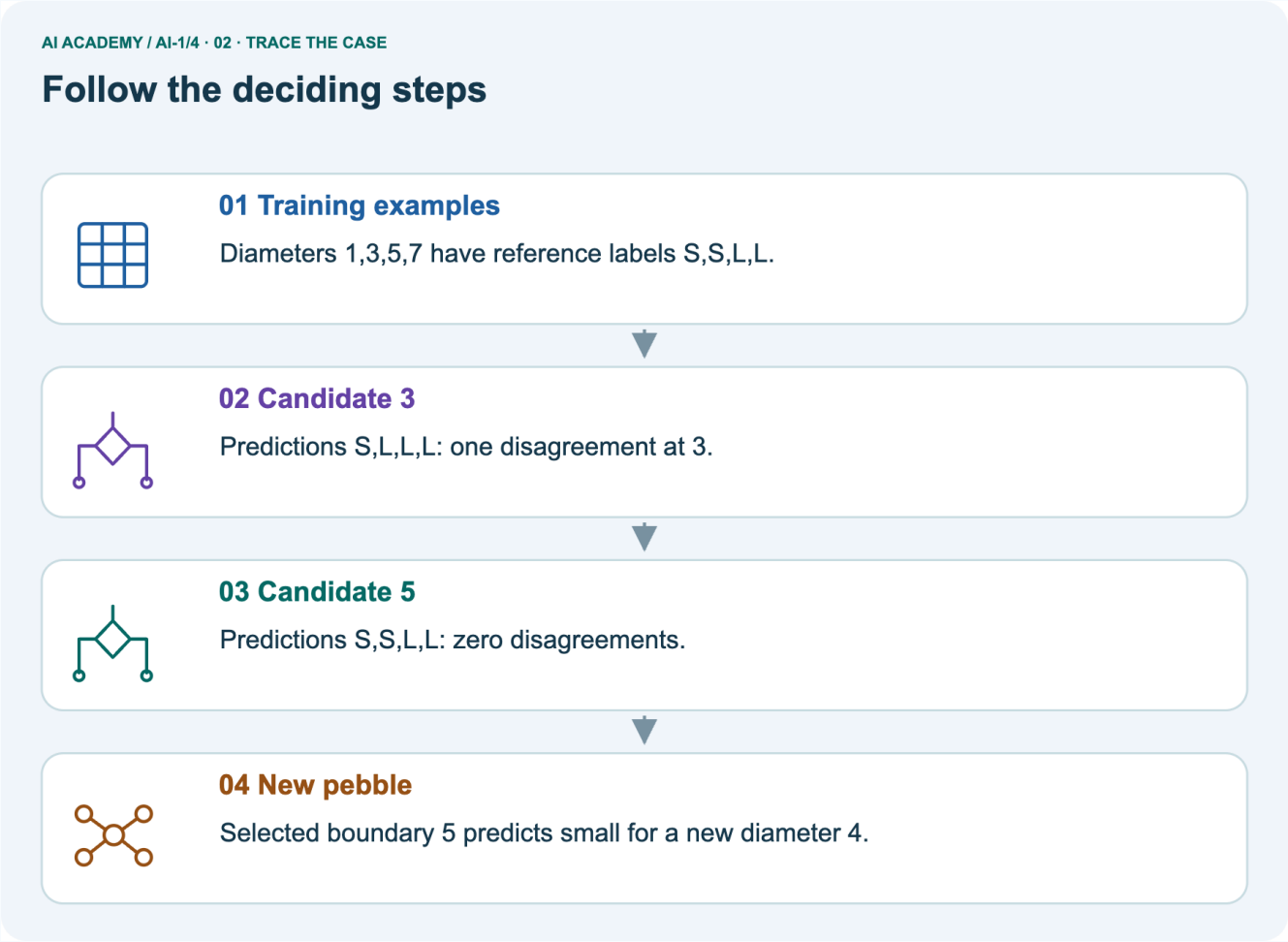
A fictional paper sorter learns a size boundary for toy pebbles. Supplied facts: **Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large. Compare large-at-least-3 with large-at-least-5; choose fewer errors.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

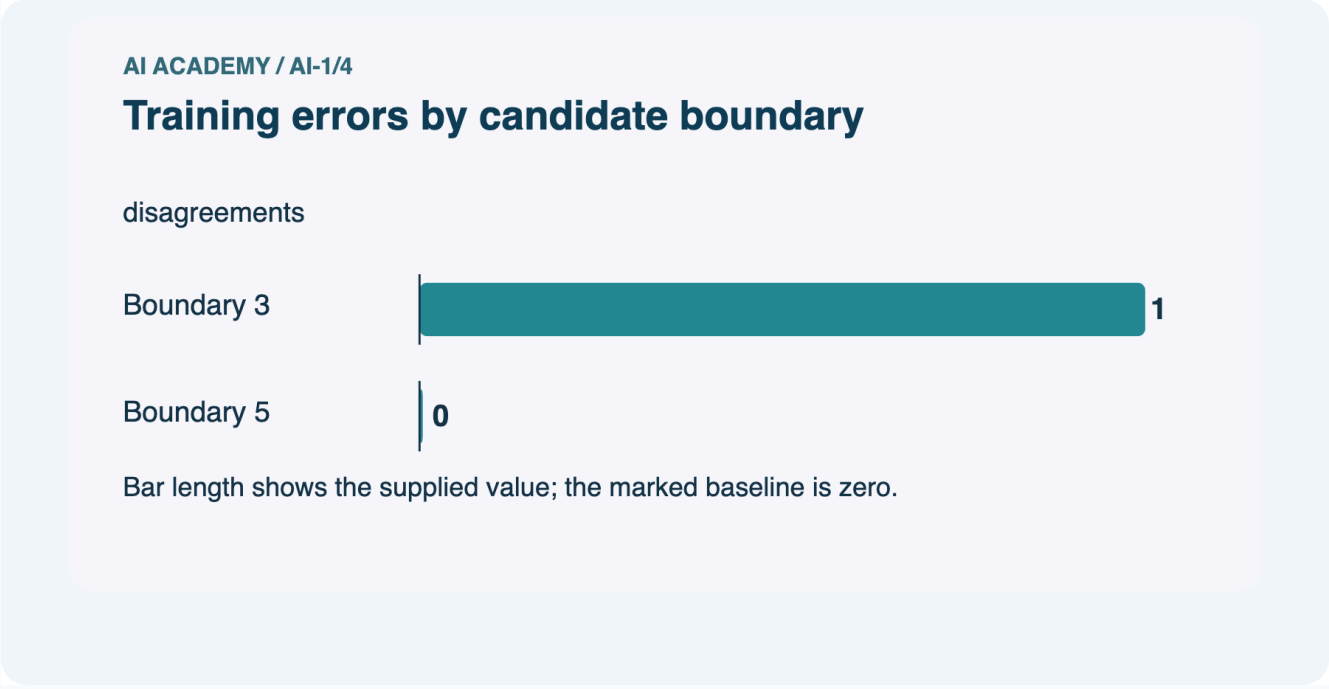
**Training** examples: Diameters 1,3,5,7 have reference **label**s S,S,L,L. → Candidate 3: **Prediction**s S,L,L,L: one disagreement at 3. → Candidate 5: **Prediction**s S,S,L,L: zero disagreements. → New pebble: Selected boundary 5 predicts small for a new diameter 4.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Training errors by candidate boundary




Follow the labelled picture with the matching explanation. Open the vector diagram

**Training** errors by candidate boundary. Units: disagreements. Boundary 3: 1. Boundary 5: 0.

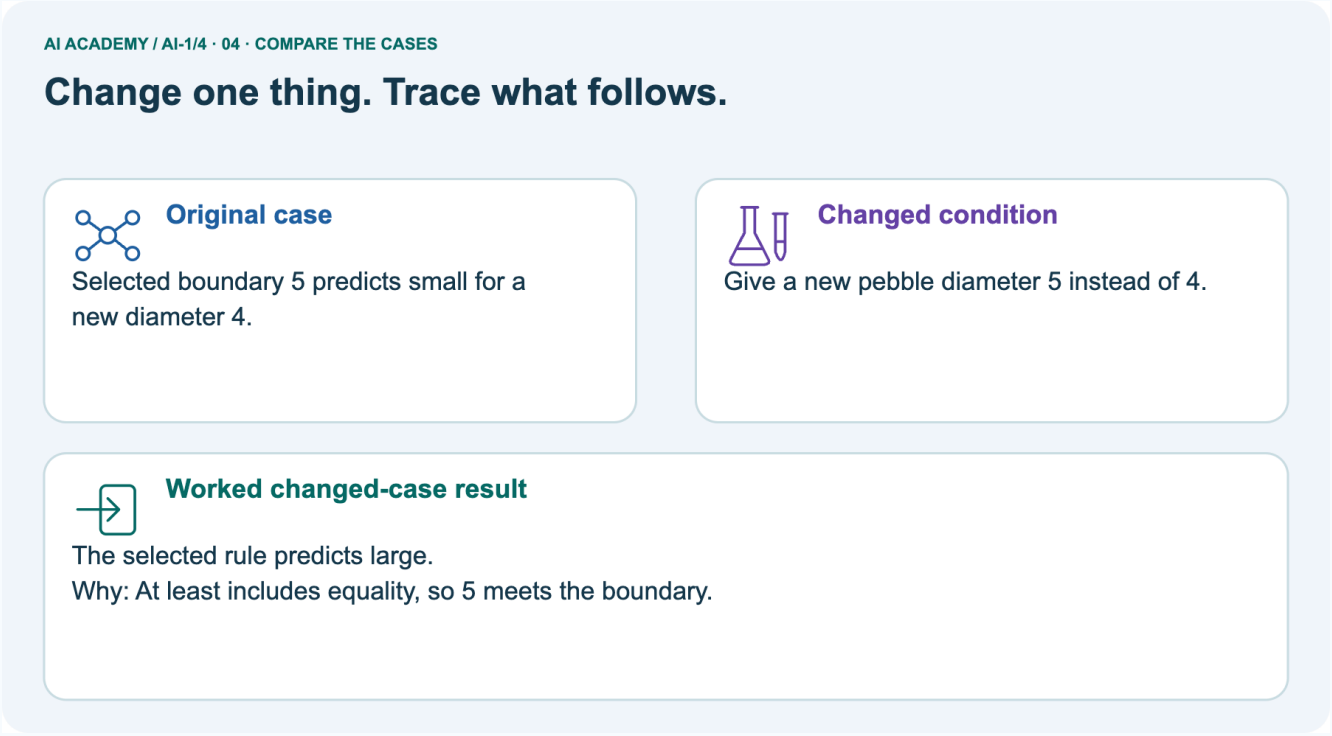
Read the **label**s and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the **label**s and values. Explain the relationship the model shows. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Selected boundary 5 predicts small for a new diameter 4. Changed condition: Give a new pebble diameter 5 instead of 4. Expected result: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

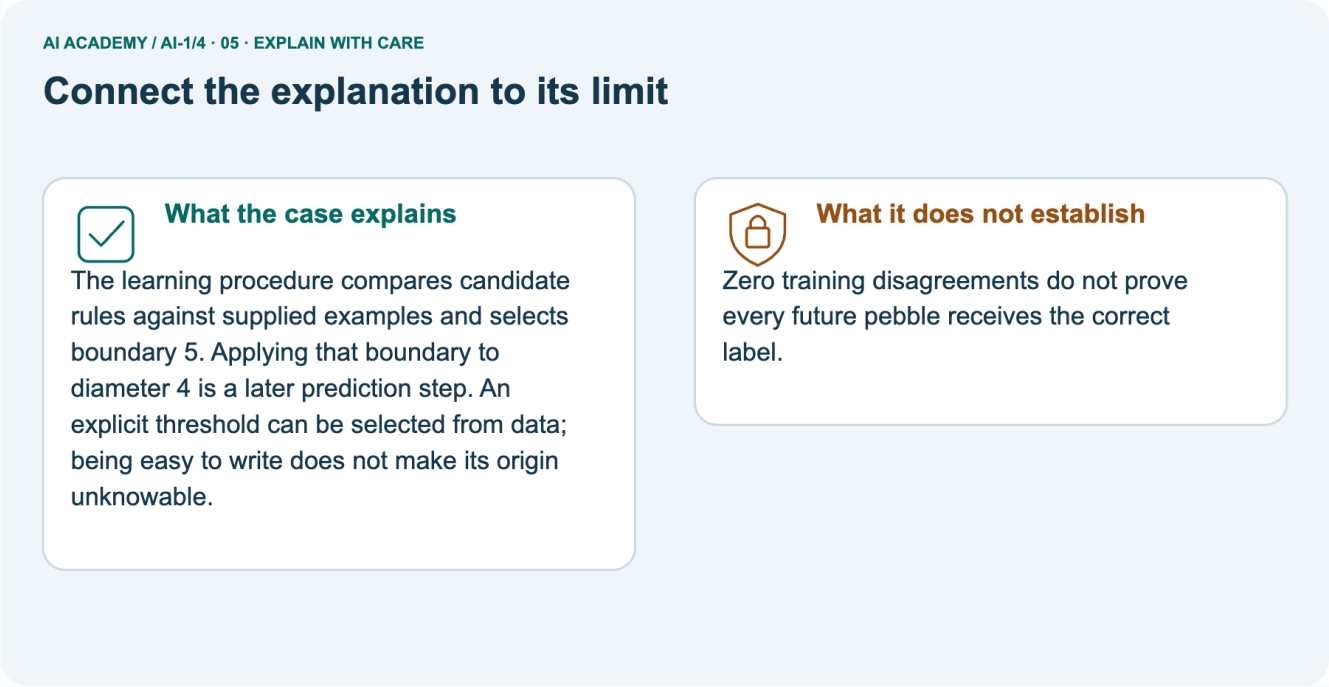
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one **label** or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: At least includes equality, so 5 meets the boundary.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable. Limit: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

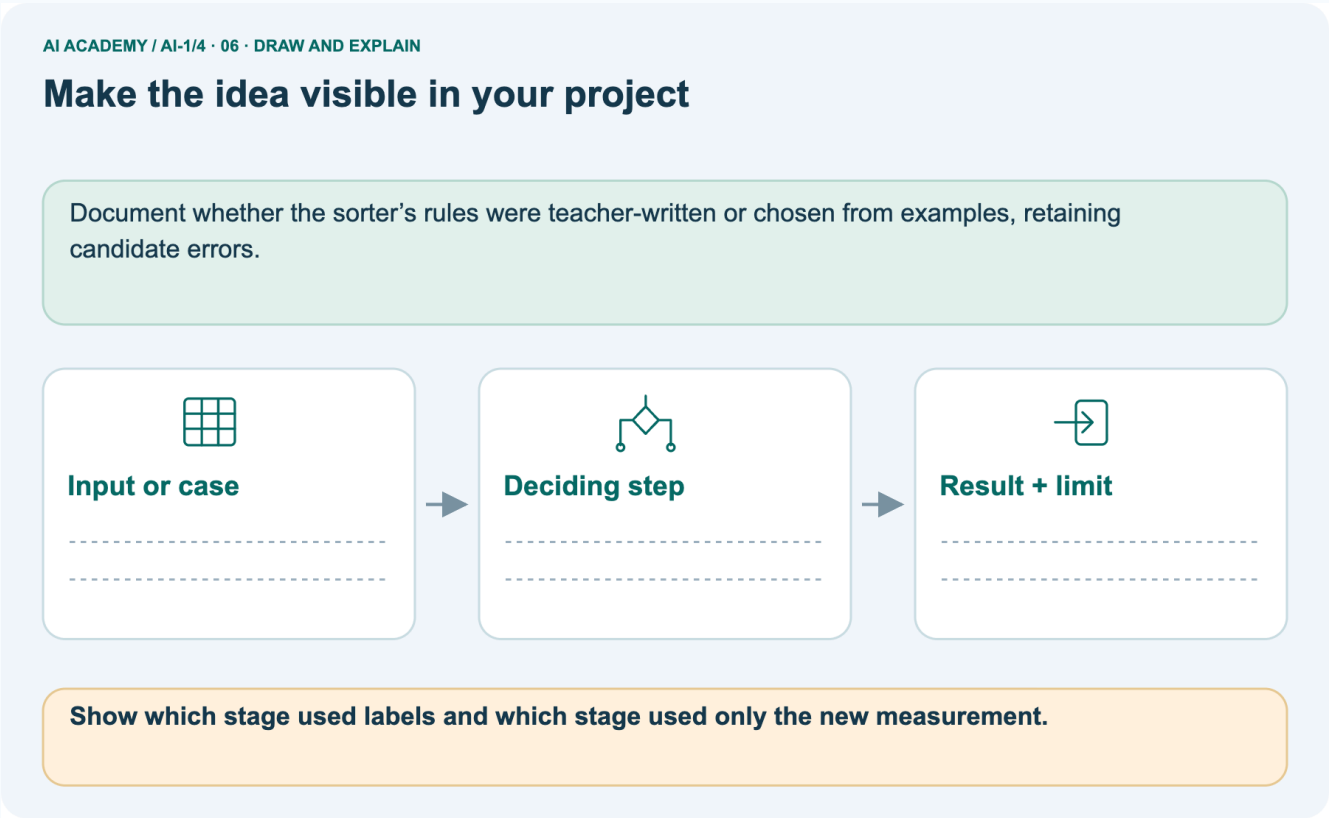
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors. Draw your **input**, deciding step, result and limit.

Supported question: Show which stage used **label**s and which stage used only the new measurement.

> Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Selection used all four **training label**s. The new **prediction** compares diameter with 5 without seeing its reference **label**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.